

# Winter Storm Fern Situation Report

January 2026

Formatted based on LA Fire [Situation Report](#)

## Summary

A major American winter storm, often referred to as Winter Storm Fern, started on Friday, January 23 and continued through January 26th. The storm brought heavy snow and freezing rain to several U.S. states, ranging from the southern plains to the Eastern United States. Several states experienced network outages and extremely cold temperatures. The following visualizations are developed based on Meta AI for Good's data through Facebook. The data covers network coverage and population density from January 27 to February 10, with a focus on Tennessee and Kentucky and daily updates during the collection period.

## Definitions

- **Baseline Ratio:** Ratio of people using the Facebook app who have turned on the Location Services device setting on their mobile device that are present in a specific tile during the baseline period over users in all tiles in the boundary box defining the crisis
- **Crisis Ratio:** Ratio of people using the Facebook app who have turned on the Location Services device setting on their mobile device that are present in a specific tile during the 8-hour window over users in all tiles in the boundary box defining the crisis
- **Population Change Density (%):** Percent change in Facebook Location Services-enabled user population counts per tile, calculated by dividing the difference by the baseline (plus a small value, usually 1)
- **Time Windows:** aggregated over 3 fixed 8-hour time windows, beginning at 00:00, 08:00 and 16:00, that represent changes in population density

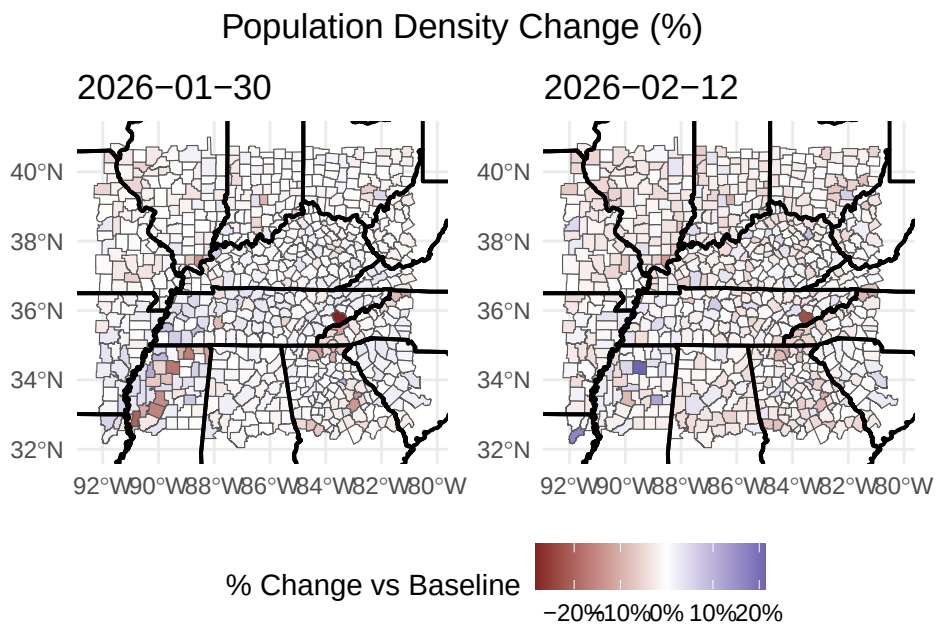
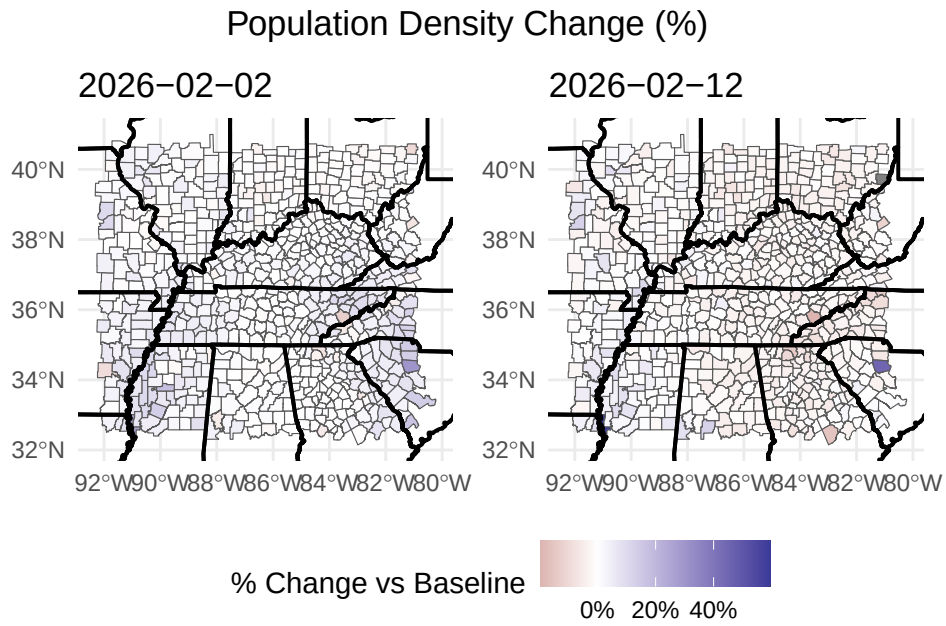
## Bing Tiles vs. Administrative Regions

The Facebook Population During Crisis (Administrative Regions) dataset is equivalent to the Facebook Population During Crisis (Tile Level) dataset except instead of aggregating user counts over grid tiles, it aggregates user counts within administrative boundaries such as states, provinces and counties.

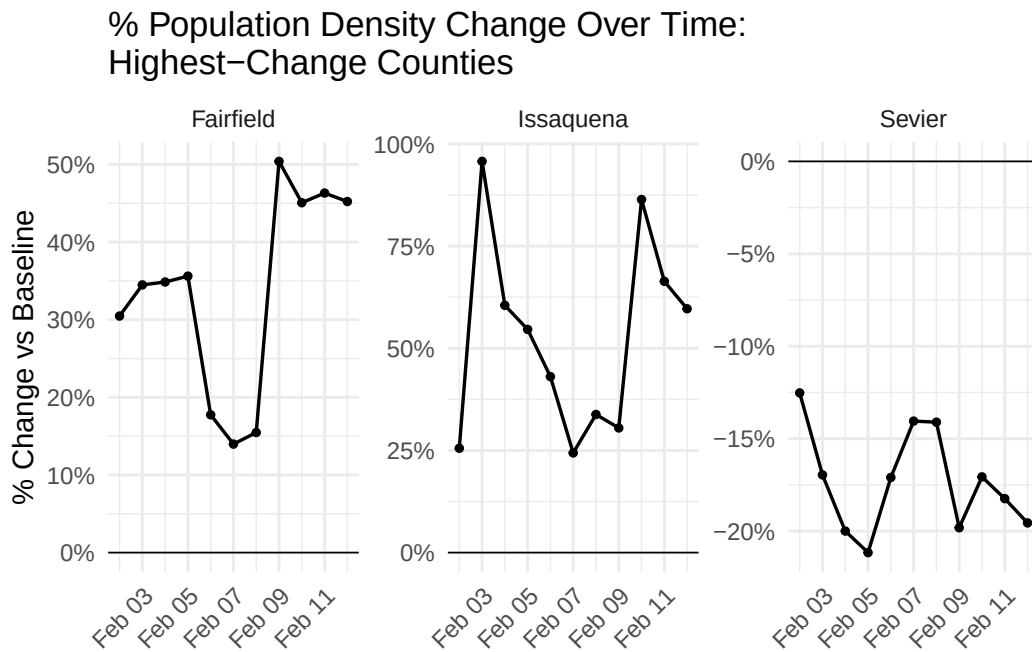
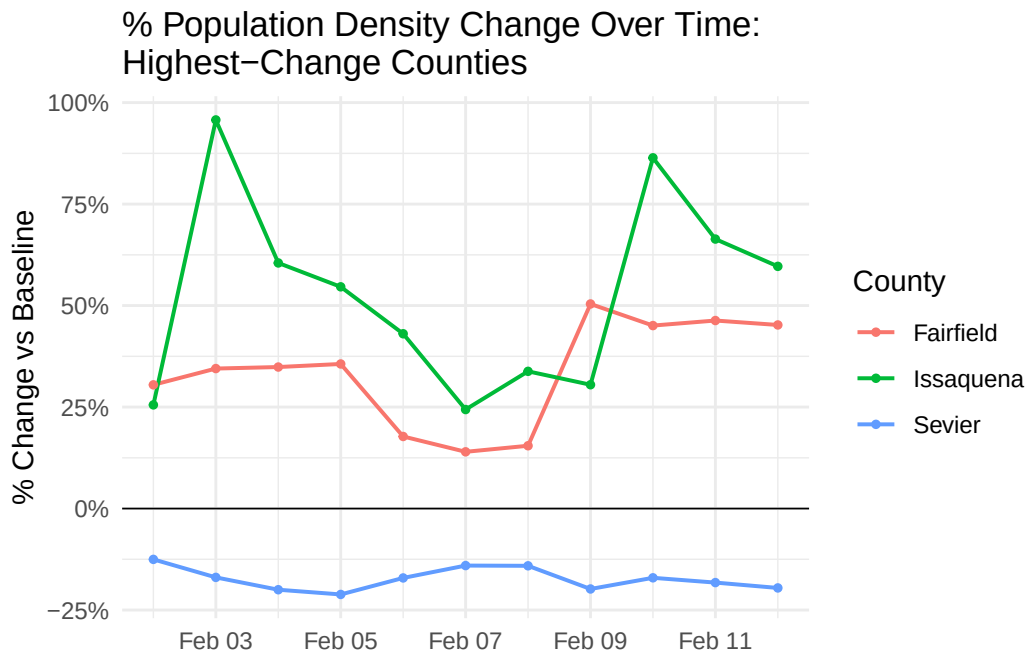
- **Bing Tiles:** equivalent to roughly 2.4 kilometers on a side near the equator or the size of 6 x 6 city blocks
- **Administrative Regions:** the polygon boundaries for administrative regions derived from the Database of Global Administrative Areas, also known as GADM.

## Movement

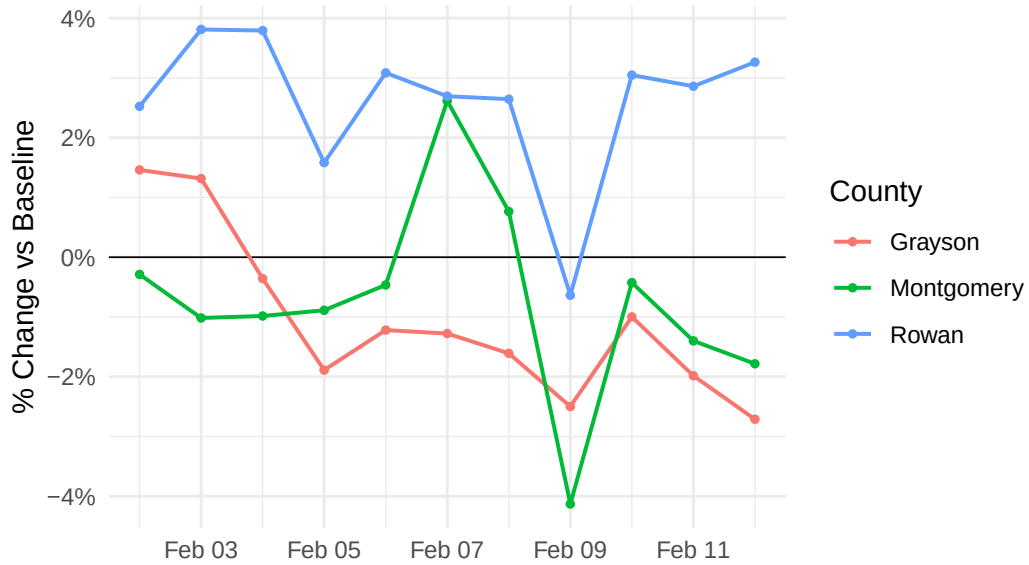
### % Population Density Change



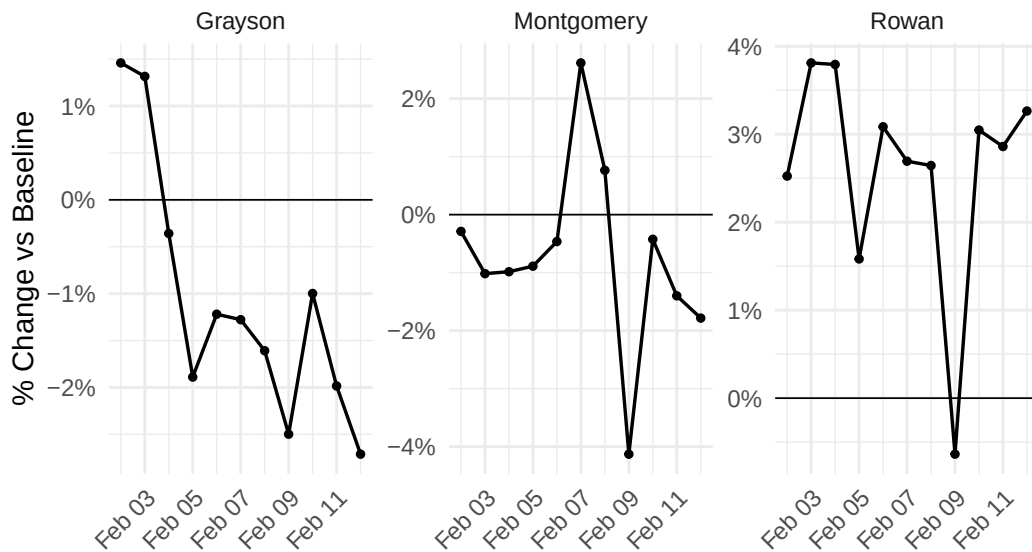
## Time Series of Movement

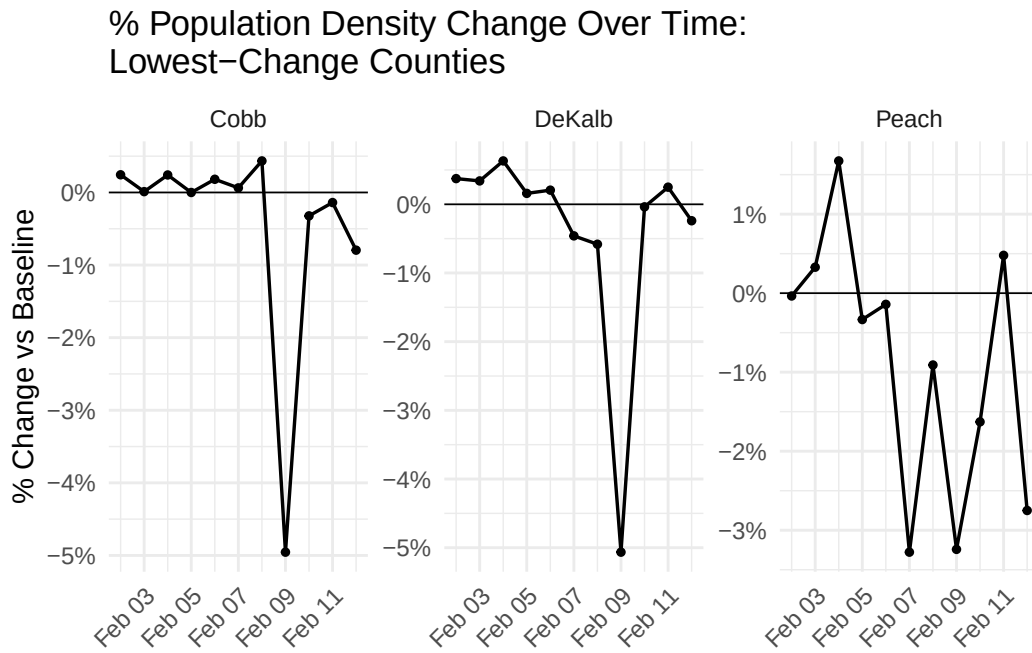
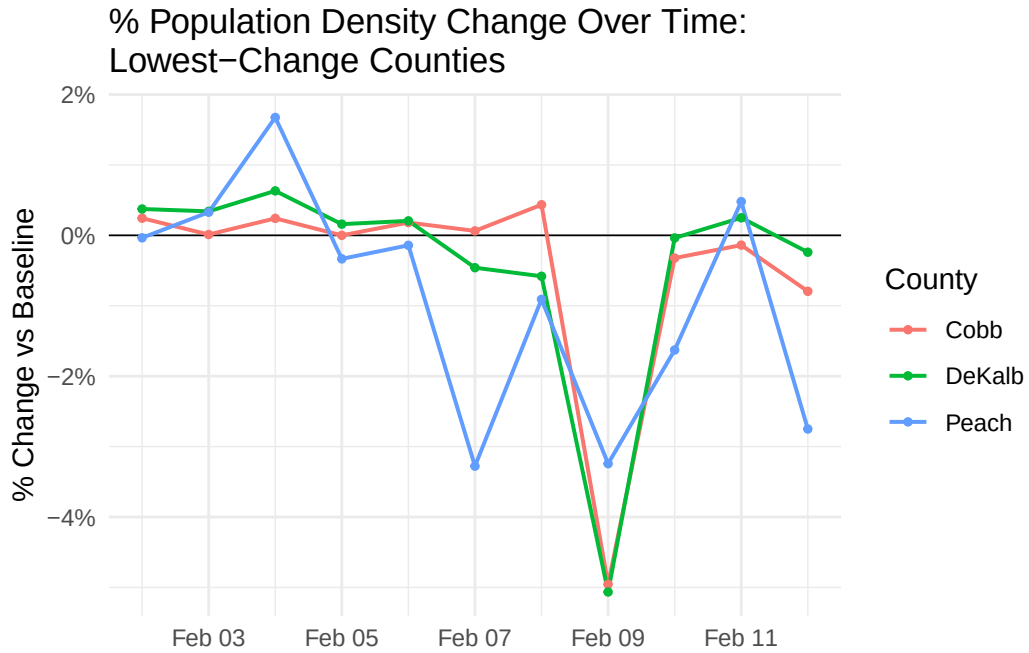


% Population Density Change Over Time:  
Average-Change Counties



% Population Density Change Over Time:  
Average-Change Counties





Places

Table 1

| start_county_name | end_county_name | total_crisis_movement | avg_z_score |
|-------------------|-----------------|-----------------------|-------------|
| Jefferson         | Jefferson       | 6548192               | 0.1003103   |
| Franklin          | Franklin        | 5262301               | -0.2757655  |
| Hamilton          | Hamilton        | 4797827               | -0.2801593  |
| Madison           | Madison         | 4333353               | -0.1587142  |
| Shelby            | Shelby          | 4228365               | -0.2163702  |
| Marion            | Marion          | 4156341               | -0.2564067  |
| St. Louis         | St. Louis       | 3805363               | -0.3254192  |
| Montgomery        | Montgomery      | 3156012               | -0.2643781  |
| Fulton            | Fulton          | 2608909               | -0.4469157  |
| Davidson          | Davidson        | 2534903               | 0.5791654   |
| Gwinnett          | Gwinnett        | 2526951               | -0.9734793  |
| Knox              | Knox            | 2223121               | 0.3169843   |
| Fayette           | Fayette         | 2059553               | -0.1049316  |
| Cobb              | Cobb            | 1921182               | -0.2391213  |
| Greenville        | Greenville      | 1897619               | 0.1128207   |
| DeKalb            | DeKalb          | 1868152               | -0.5146540  |
| Warren            | Warren          | 1585751               | -0.4665952  |
| Butler            | Butler          | 1498309               | -0.1566856  |
| Rutherford        | Rutherford      | 1473273               | -0.4580661  |
| Richland          | Richland        | 1442676               | 0.5490029   |

## Counties with Most Increased Population Density on 2026-02-02

Source: Bing Tiles

| County     | State          | Population<br>Density<br>Change | Total<br>Population | Median<br>Household<br>Income | Poverty<br>Rate | % Age<br>65+ | % No<br>Vehicle<br>Access |
|------------|----------------|---------------------------------|---------------------|-------------------------------|-----------------|--------------|---------------------------|
| Fairfield  | South Carolina | 30.5                            | 20,550              | \$47,885                      | 15.6%           | 3.2%         | 5.9%                      |
| Issaquena  | Mississippi    | 25.5                            | 928                 | \$31,429                      | 16.2%           | 2.7%         | 14.0%                     |
| Grenada    | Mississippi    | 17.9                            | 21,217              | \$48,804                      | 24.6%           | 2.3%         | 7.9%                      |
| Hampton    | South Carolina | 16.6                            | 18,254              | \$44,711                      | 20.7%           | 2.2%         | 11.0%                     |
| York       | South Carolina | 14.5                            | 293,673             | \$89,095                      | 8.4%            | 2.2%         | 3.7%                      |
| Orangeburg | South Carolina | 13.8                            | 83,253              | \$45,675                      | 22.4%           | 3.1%         | 9.0%                      |
| Chester    | South Carolina | 13.8                            | 32,182              | \$52,458                      | 18.7%           | 2.5%         | 6.3%                      |
| Holmes     | Mississippi    | 13.5                            | 16,191              | \$32,538                      | 36.4%           | 1.9%         | 15.9%                     |
| Lincoln    | North Carolina | 12.7                            | 92,716              | \$80,016                      | 11.2%           | 2.9%         | 3.3%                      |
| Lake       | Tennessee      | 12.5                            | 6,579               | \$28,814                      | 32.2%           | 2.2%         | 18.3%                     |
| Alexander  | North Carolina | 12.2                            | 36,412              | \$65,354                      | 12.6%           | 2.4%         | 3.1%                      |
| Quitman    | Mississippi    | 11.0                            | 5,774               | \$32,412                      | 29.6%           | 2.4%         | 10.9%                     |
| Alexander  | Illinois       | 10.7                            | 4,875               | \$47,043                      | 19.0%           | 4.4%         | 10.1%                     |
| Gaston     | North Carolina | 10.7                            | 234,881             | \$67,478                      | 13.5%           | 2.3%         | 5.1%                      |
| Osage      | Missouri       | 10.6                            | 13,402              | \$76,681                      | 8.6%            | 2.6%         | 5.6%                      |

## Counties with Most Decreased Population Density on 2026-02-02

Source: Bing Tiles

| County     | State          | Population<br>Density<br>Change | Total<br>Population | Median<br>Household<br>Income | Poverty<br>Rate | % Age<br>65+ | % No<br>Vehicle<br>Access |
|------------|----------------|---------------------------------|---------------------|-------------------------------|-----------------|--------------|---------------------------|
| Sevier     | Tennessee      | -12.5                           | 99,652              | \$62,581                      | 13.5%           | 2.7%         | 4.1%                      |
| Carroll    | Ohio           | -9.6                            | 26,659              | \$64,835                      | 12.9%           | 3.5%         | 6.8%                      |
| Jefferson  | Arkansas       | -8.9                            | 64,802              | \$51,096                      | 18.6%           | 2.8%         | 7.5%                      |
| Clay       | West Virginia  | -6.7                            | 7,835               | \$42,318                      | 26.4%           | 3.5%         | 10.3%                     |
| Fannin     | Georgia        | -6.7                            | 25,742              | \$57,073                      | 13.7%           | 2.9%         | 3.0%                      |
| Taliaferro | Georgia        | -6.1                            | 1,558               | \$52,500                      | 20.4%           | 2.9%         | 5.8%                      |
| Brown      | Indiana        | -5.4                            | 15,606              | \$78,528                      | 7.8%            | 4.5%         | 3.5%                      |
| Swain      | North Carolina | -5.4                            | 14,013              | \$52,349                      | 19.1%           | 3.8%         | 6.6%                      |
| Taylor     | Georgia        | -5.2                            | 7,786               | \$41,788                      | 32.1%           | 4.2%         | 15.5%                     |
| Greene     | Alabama        | -4.8                            | 7,424               | \$29,200                      | 40.1%           | 4.5%         | 13.3%                     |
| Watauga    | North Carolina | -4.5                            | 55,123              | \$51,693                      | 25.5%           | 2.3%         | 4.4%                      |
| White      | Georgia        | -3.6                            | 28,810              | \$70,666                      | 12.4%           | 3.2%         | 2.6%                      |
| Gilmer     | Georgia        | -3.4                            | 32,426              | \$74,499                      | 16.0%           | 3.2%         | 2.9%                      |
| Tishomingo | Mississippi    | -3.3                            | 18,660              | \$53,040                      | 18.0%           | 2.7%         | 4.8%                      |
| Morgan     | Indiana        | -3.3                            | 72,659              | \$79,429                      | 9.3%            | 3.2%         | 3.6%                      |

## Demographic Profile



## Counties with Most Increased Population Density on 2026-02-12

Source: Bing Tiles

| County       | State          | Population<br>Density<br>Change | Total<br>Population | Median<br>Household<br>Income | Poverty<br>Rate | % Age<br>65+ | % No<br>Vehicle<br>Access |
|--------------|----------------|---------------------------------|---------------------|-------------------------------|-----------------|--------------|---------------------------|
| Issaquena    | Mississippi    | 59.7                            | 928                 | \$31,429                      | 16.2%           | 2.7%         | 14.0%                     |
| Fairfield    | South Carolina | 45.2                            | 20,550              | \$47,885                      | 15.6%           | 3.2%         | 5.9%                      |
| Osage        | Missouri       | 12.6                            | 13,402              | \$76,681                      | 8.6%            | 2.6%         | 5.6%                      |
| Perry        | Alabama        | 12.6                            | 8,031               | \$37,654                      | 31.4%           | 3.3%         | 8.0%                      |
| New Madrid   | Missouri       | 10.6                            | 15,731              | \$51,881                      | 17.7%           | 3.3%         | 7.2%                      |
| Norton       | Virginia       | 10.1                            | 3,577               | \$41,495                      | 26.3%           | 2.3%         | 10.7%                     |
| Brown        | Illinois       | 9.9                             | 6,322               | \$67,917                      | 9.2%            | 3.1%         | 8.5%                      |
| Humphreys    | Mississippi    | 9.5                             | 7,395               | \$33,731                      | 27.0%           | 3.1%         | 13.0%                     |
| Coahoma      | Mississippi    | 9.3                             | 20,540              | \$37,461                      | 35.5%           | 2.3%         | 14.9%                     |
| East Carroll | Louisiana      | 8.9                             | 7,072               | \$33,341                      | 35.2%           | 1.3%         | 9.2%                      |
| Sumter       | Alabama        | 8.4                             | 11,878              | \$33,310                      | 26.5%           | 2.7%         | 9.5%                      |
| Leflore      | Mississippi    | 7.9                             | 27,141              | \$35,277                      | 31.5%           | 2.1%         | 14.7%                     |
| Holmes       | Mississippi    | 7.8                             | 16,191              | \$32,538                      | 36.4%           | 1.9%         | 15.9%                     |
| Lee          | Arkansas       | 7.6                             | 8,359               | \$34,375                      | 39.4%           | 2.3%         | 16.7%                     |
| Reynolds     | Missouri       | 7.4                             | 6,010               | \$45,028                      | 17.3%           | 3.8%         | 7.4%                      |

## Counties with Most Decreased Population Density on 2026-02-12

Source: Bing Tiles

| County    | State          | Population<br>Density<br>Change | Total<br>Population | Median<br>Household<br>Income | Poverty<br>Rate | % Age<br>65+ | % No<br>Vehicle<br>Access |
|-----------|----------------|---------------------------------|---------------------|-------------------------------|-----------------|--------------|---------------------------|
| Sevier    | Tennessee      | -19.6                           | 99,652              | \$62,581                      | 13.5%           | 2.7%         | 4.1%                      |
| Laurens   | Georgia        | -15.6                           | 49,798              | \$55,010                      | 22.6%           | 2.3%         | 6.3%                      |
| Fannin    | Georgia        | -13.8                           | 25,742              | \$57,073                      | 13.7%           | 2.9%         | 3.0%                      |
| Alleghany | North Carolina | -13.8                           | 11,174              | \$47,172                      | 16.0%           | 2.6%         | 6.0%                      |
| Clay      | West Virginia  | -11.3                           | 7,835               | \$42,318                      | 26.4%           | 3.5%         | 10.3%                     |
| Gilmer    | Georgia        | -10.1                           | 32,426              | \$74,499                      | 16.0%           | 3.2%         | 2.9%                      |
| Ashe      | North Carolina | -9.1                            | 26,950              | \$56,866                      | 13.5%           | 3.5%         | 4.4%                      |
| Wilkes    | North Carolina | -8.7                            | 65,935              | \$51,721                      | 16.7%           | 2.6%         | 5.8%                      |
| White     | Georgia        | -8.6                            | 28,810              | \$70,666                      | 12.4%           | 3.2%         | 2.6%                      |
| Watauga   | North Carolina | -8.5                            | 55,123              | \$51,693                      | 25.5%           | 2.3%         | 4.4%                      |
| Macon     | North Carolina | -8.1                            | 37,941              | \$59,061                      | 11.6%           | 3.2%         | 6.1%                      |
| Brown     | Indiana        | -7.3                            | 15,606              | \$78,528                      | 7.8%            | 4.5%         | 3.5%                      |
| Taylor    | Georgia        | -7.1                            | 7,786               | \$41,788                      | 32.1%           | 4.2%         | 15.5%                     |
| Hocking   | Ohio           | -7.0                            | 27,799              | \$62,960                      | 16.0%           | 3.3%         | 5.7%                      |
| Morgan    | Indiana        | -6.8                            | 72,659              | \$79,429                      | 9.3%            | 3.2%         | 3.6%                      |

## Counties with Most Increased Population Density on 2026-01-30

Source: Administrative Regions

| County    | State       | Population<br>Density<br>Change | Total<br>Population | Median<br>Household<br>Income | Poverty<br>Rate | % Age<br>65+ | % No<br>Vehicle<br>Access |
|-----------|-------------|---------------------------------|---------------------|-------------------------------|-----------------|--------------|---------------------------|
| Quitman   | Mississippi | 11.6                            | 5,774               | \$32,412                      | 29.6%           | 2.4%         | 10.9%                     |
| Clarke    | Georgia     | 10.6                            | 129,609             | \$53,625                      | 26.0%           | 1.5%         | 6.3%                      |
| Tate      | Mississippi | 9.3                             | 28,321              | \$69,704                      | 16.9%           | 2.7%         | 3.0%                      |
| Chester   | Tennessee   | 9.1                             | 17,611              | \$59,341                      | 16.1%           | 1.2%         | 3.1%                      |
| Tipton    | Tennessee   | 8.1                             | 61,553              | \$74,127                      | 13.6%           | 2.4%         | 5.2%                      |
| Oktober   | Mississippi | 7.4                             | 51,771              | \$46,695                      | 25.5%           | 1.7%         | 5.4%                      |
| Fayette   | Tennessee   | 6.5                             | 43,267              | \$88,456                      | 10.3%           | 3.5%         | 5.3%                      |
| Union     | Kentucky    | 6.4                             | 13,260              | \$60,327                      | 18.4%           | 3.0%         | 5.6%                      |
| Lee       | Mississippi | 6.2                             | 83,034              | \$67,863                      | 14.6%           | 2.1%         | 6.0%                      |
| Chicot    | Arkansas    | 5.9                             | 9,765               | \$39,045                      | 22.6%           | 3.6%         | 10.1%                     |
| Fulton    | Kentucky    | 5.8                             | 6,409               | \$38,217                      | 22.6%           | 2.5%         | 17.0%                     |
| Morehouse | Louisiana   | 5.6                             | 24,551              | \$39,454                      | 29.3%           | 3.6%         | 11.6%                     |
| Athens    | Ohio        | 5.4                             | 61,737              | \$56,001                      | 21.5%           | 1.7%         | 6.8%                      |
| Weakley   | Tennessee   | 5.0                             | 33,016              | \$51,880                      | 19.0%           | 2.6%         | 5.5%                      |
| Monroe    | Arkansas    | 4.9                             | 6,589               | \$43,214                      | 20.6%           | 3.1%         | 14.5%                     |

## Counties with Most Decreased Population Density on 2026-01-30

Source: Administrative Regions

| County     | State          | Population<br>Density<br>Change | Total<br>Population | Median<br>Household<br>Income | Poverty<br>Rate | % Age<br>65+ | % No<br>Vehicle<br>Access |
|------------|----------------|---------------------------------|---------------------|-------------------------------|-----------------|--------------|---------------------------|
| Sevier     | Tennessee      | -28.3                           | 99,652              | \$62,581                      | 13.5%           | 2.7%         | 4.1%                      |
| Sharkey    | Mississippi    | -20.3                           | 3,872               | \$40,000                      | 30.9%           | 2.1%         | 8.0%                      |
| Lafayette  | Mississippi    | -17.5                           | 58,327              | \$67,185                      | 18.5%           | 2.0%         | 6.7%                      |
| Tippah     | Mississippi    | -16.6                           | 21,523              | \$52,542                      | 18.9%           | 3.6%         | 2.2%                      |
| Holmes     | Mississippi    | -15.9                           | 16,191              | \$32,538                      | 36.4%           | 1.9%         | 15.9%                     |
| Carroll    | Mississippi    | -14.7                           | 9,686               | \$59,327                      | 17.9%           | 1.7%         | 4.5%                      |
| Pope       | Illinois       | -13.6                           | 3,739               | \$60,050                      | 13.3%           | 4.1%         | 5.5%                      |
| Hancock    | Georgia        | -13.4                           | 8,650               | \$40,082                      | 30.6%           | 3.6%         | 8.4%                      |
| Taliaferro | Georgia        | -13.2                           | 1,558               | \$52,500                      | 20.4%           | 2.9%         | 5.8%                      |
| Swain      | North Carolina | -12.7                           | 14,013              | \$52,349                      | 19.1%           | 3.8%         | 6.6%                      |
| Fannin     | Georgia        | -12.4                           | 25,742              | \$57,073                      | 13.7%           | 2.9%         | 3.0%                      |
| Alleghany  | North Carolina | -12.1                           | 11,174              | \$47,172                      | 16.0%           | 2.6%         | 6.0%                      |
| Tishomingo | Mississippi    | -11.7                           | 18,660              | \$53,040                      | 18.0%           | 2.7%         | 4.8%                      |
| Humphreys  | Mississippi    | -10.8                           | 7,395               | \$33,731                      | 27.0%           | 3.1%         | 13.0%                     |
| White      | Georgia        | -10.0                           | 28,810              | \$70,666                      | 12.4%           | 3.2%         | 2.6%                      |

## Counties with Most Increased Population Density on 2026-02-12

Source: Administrative Regions

| County    | State          | Population<br>Density<br>Change | Total<br>Population | Median<br>Household<br>Income | Poverty<br>Rate | % Age<br>65+ | % No<br>Vehicle<br>Access |
|-----------|----------------|---------------------------------|---------------------|-------------------------------|-----------------|--------------|---------------------------|
| Lafayette | Mississippi    | 21.4                            | 58,327              | \$67,185                      | 18.5%           | 2.0%         | 6.7%                      |
| Richland  | Louisiana      | 16.7                            | 19,822              | \$53,544                      | 24.0%           | 2.9%         | 8.7%                      |
| Oktibbeha | Mississippi    | 12.2                            | 51,771              | \$46,695                      | 25.5%           | 1.7%         | 5.4%                      |
| Norton    | Virginia       | 9.7                             | 3,577               | \$41,495                      | 26.3%           | 2.3%         | 10.7%                     |
| Clarke    | Georgia        | 9.5                             | 129,609             | \$53,625                      | 26.0%           | 1.5%         | 6.3%                      |
| Rowan     | Kentucky       | 8.2                             | 24,578              | \$54,321                      | 24.2%           | 1.3%         | 8.4%                      |
| Davidson  | Tennessee      | 6.9                             | 715,388             | \$77,853                      | 13.9%           | 1.9%         | 6.5%                      |
| Athens    | Ohio           | 6.5                             | 61,737              | \$56,001                      | 21.5%           | 1.7%         | 6.8%                      |
| Grenada   | Mississippi    | 5.5                             | 21,217              | \$48,804                      | 24.6%           | 2.3%         | 7.9%                      |
| Panola    | Mississippi    | 5.2                             | 32,808              | \$45,945                      | 23.3%           | 2.1%         | 6.4%                      |
| Osage     | Missouri       | 5.1                             | 13,402              | \$76,681                      | 8.6%            | 2.6%         | 5.6%                      |
| Richland  | South Carolina | 4.9                             | 422,117             | \$63,784                      | 17.2%           | 2.2%         | 7.2%                      |
| Christian | Kentucky       | 4.9                             | 72,069              | \$55,494                      | 17.1%           | 1.8%         | 7.8%                      |
| Fulton    | Georgia        | 4.9                             | 1,076,561           | \$95,292                      | 12.3%           | 1.8%         | 10.0%                     |
| Fayette   | Kentucky       | 4.8                             | 323,725             | \$69,479                      | 14.9%           | 2.2%         | 7.2%                      |

## Counties with Most Decreased Population Density on 2026-02-12

Source: Administrative Regions

| County     | State          | Population<br>Density<br>Change | Total<br>Population | Median<br>Household<br>Income | Poverty<br>Rate | % Age<br>65+ | % No<br>Vehicle<br>Access |
|------------|----------------|---------------------------------|---------------------|-------------------------------|-----------------|--------------|---------------------------|
| Sevier     | Tennessee      | -22.8                           | 99,652              | \$62,581                      | 13.5%           | 2.7%         | 4.1%                      |
| Pope       | Illinois       | -12.5                           | 3,739               | \$60,050                      | 13.3%           | 4.1%         | 5.5%                      |
| Alleghany  | North Carolina | -11.3                           | 11,174              | \$47,172                      | 16.0%           | 2.6%         | 6.0%                      |
| Fannin     | Georgia        | -11.3                           | 25,742              | \$57,073                      | 13.7%           | 2.9%         | 3.0%                      |
| Calhoun    | Illinois       | -11.1                           | 4,330               | \$93,203                      | 6.3%            | 2.7%         | 2.7%                      |
| Taliaferro | Georgia        | -10.6                           | 1,558               | \$52,500                      | 20.4%           | 2.9%         | 5.8%                      |
| Brown      | Indiana        | -9.9                            | 15,606              | \$78,528                      | 7.8%            | 4.5%         | 3.5%                      |
| Carroll    | Mississippi    | -9.0                            | 9,686               | \$59,327                      | 17.9%           | 1.7%         | 4.5%                      |
| Gilmer     | Georgia        | -8.7                            | 32,426              | \$74,499                      | 16.0%           | 3.2%         | 2.9%                      |
| Ashe       | North Carolina | -8.7                            | 26,950              | \$56,866                      | 13.5%           | 3.5%         | 4.4%                      |
| Hancock    | Georgia        | -8.4                            | 8,650               | \$40,082                      | 30.6%           | 3.6%         | 8.4%                      |
| Talbot     | Georgia        | -8.2                            | 5,742               | \$41,597                      | 25.2%           | 4.0%         | 9.6%                      |
| White      | Georgia        | -8.1                            | 28,810              | \$70,666                      | 12.4%           | 3.2%         | 2.6%                      |
| Noble      | Ohio           | -7.9                            | 14,282              | \$56,098                      | 10.6%           | 3.3%         | 7.7%                      |
| Harris     | Georgia        | -7.4                            | 36,086              | \$97,302                      | 9.0%            | 2.8%         | 1.4%                      |